SAP S/4HANA SECURITY

Chapter 2 – SAP Fiori Security (Complete Architect Guide)

Q121. What is SAP Fiori Security?

Interviewer's Objective

The interviewer wants to understand whether you know that Fiori Security consists of multiple layers rather than simply assigning catalogs.

Executive Answer

SAP Fiori Security is the authorization framework that controls user access to SAP Fiori applications, Launchpad content, OData services, backend business transactions, and business data.

Unlike SAP GUI, where access is primarily transaction-based, Fiori introduces multiple security layers that work together.

These layers include:

- **Launchpad Security**
- **Catalog Security**
- **Space & Page Security**
- **Target Mapping**
- **OData Service Authorization**
- **Backend Role Authorization**
- **CDS Authorization**
- **Business Data Authorization**

A user must successfully pass all applicable layers before a Fiori application functions correctly.

Fiori Security Layers

User
│
Authentication (IAS / SSO / Entra ID)

│

Launchpad Login

│

Catalog Authorization

│

Target Mapping

│

Space & Page Assignment

│

OData Service

│

Backend Authorization

│

Business Data Authorization

│

Application Opens

Q122. Explain the Fiori Security Architecture.

Executive Answer

A complete Fiori Security architecture consists of two major components.

Frontend Server (FES)
│
▼
Launchpad
Catalogs
Spaces
Pages
Target Mapping
OData Services
│
▼
Backend Server (BES)
│
Authorization Objects
Business Roles
Organizational Levels
Business Data

Frontend Responsibilities

- **Launchpad**
- **Tiles**
- **Catalogs**
- **Spaces**
- **Pages**
- **Target Mappings**
- **UI Access**

Backend Responsibilities

- **Authorization Objects**
- **Business Transactions**
- **Company Code Restrictions**
- **Plant Restrictions**
- **Business Data Security**

Interview Tip

Always explain Frontend and Backend separately.

Most candidates forget this.

Q123. Explain Catalogs.

One of the most common interview questions.

Executive Answer

A Catalog is a collection of applications that a user is authorized to launch.

A Catalog contains:

- **Tiles**

- **Target Mappings**
- **OData references**
- **App Launcher Information**

Think of it as the authorization container for Fiori applications.

Example

Catalog

Procurement Catalog

Contains

- **Create Purchase Order**
- **Display Purchase Order**
- **Approve Purchase Order**

Best Practice

Design catalogs based on business functions, not departments.

Q124. Explain Groups, Spaces and Pages.

Interviewers frequently ask this.

Groups (Legacy)

Older Launchpad organization.

Static collection of tiles.

SAP now recommends using Spaces and Pages.

Spaces

Represent major business areas.

Examples

- **Finance**

- **Procurement**
- **Sales**
- **Manufacturing**

Pages

Pages organize applications within a Space.

Example

Space

Finance

↓

Page

Accounts Payable

↓

Tiles

Create Invoice

Display Invoice

Approve Invoice

Executive Answer

Spaces provide high-level navigation.

Pages organize applications within each business area.

Catalogs determine authorization, while Spaces and Pages determine presentation.

Q125. What is Target Mapping?

Executive Answer

Target Mapping connects a Launchpad tile to the application that should be executed.

Without Target Mapping:

- **Tile is visible**
- **Application cannot open**

Target Mapping defines:

- **Semantic Object**
- **Action**
- **Navigation Target**

Example

Semantic Object

PurchaseOrder

↓

Action

Create

↓

Application

Create Purchase Order App

Q126. Explain OData Services.

One of the most important Fiori interview questions.

Executive Answer

OData (Open Data Protocol) is the communication layer between the Fiori application and the backend SAP system.

Instead of directly calling SAP transactions,

Fiori Apps communicate using:

Browser

↓

HTTP

↓

OData Service

↓

Backend SAP

↓

Database

Responsibilities

- **Read Data**
- **Update Data**
- **Delete Data**
- **Create Business Objects**

Interview Tip

Almost every Fiori application depends on one or more OData services.

Q127. What is the difference between Frontend and Backend Roles?

Very common interview question.

Frontend Role

Contains

- **Catalogs**
- **Spaces**
- **Pages**
- **Target Mapping**
- **Launchpad Access**

Responsible for

"What users see."

Backend Role

Contains

- **Authorization Objects**
- **Organizational Levels**
- **Business Authorizations**

Responsible for

"What users can do."

Simple Diagram

Frontend

Visible Tile

↓

Backend

Authorized Data

↓

Business Process

Q128. Explain Fiori Launchpad.

Executive Answer

The SAP Fiori Launchpad is the central entry point for all Fiori applications.

It provides:

- **Single entry point**
- **Personalized experience**
- **Role-based navigation**
- **Search**
- **Notifications**
- **Home page**
- **Spaces**
- **Pages**

The Launchpad itself does not grant authorization.

It displays applications based on assigned catalogs and spaces.

Q129. Explain ICF Nodes.

This is frequently asked by technical interviewers.

Executive Answer

ICF (Internet Communication Framework) Nodes enable web communication between SAP and browsers.

Without active ICF services,

Fiori applications cannot execute.

Transaction

SICF

is used to activate or manage ICF services.

Common Issue

Tile visible

↓

App fails

↓

Inactive ICF Node

Q130. Explain the App Index.

Executive Answer

The App Index is an internal SAP repository that maps Fiori applications, target mappings, and navigation information.

If the App Index becomes inconsistent,

users may experience:

- **Missing applications**
- **Incorrect navigation**
- **Tile errors**

After transports or upgrades, rebuilding the App Index may be necessary to synchronize metadata.

Q131. Fiori Troubleshooting Scenario

Interviewer

"The user can see the tile, but clicking it returns 'You are not authorized.' How do you troubleshoot?"

Executive Answer

I follow a structured troubleshooting process:

Step 1

Confirm whether the tile appears in the Launchpad.

If not,

check:

- **Catalog assignment**
- **Space/Page assignment**
- **Target Mapping**

Step 2

If the tile is visible,

verify:

- **OData Service activation**
- **ICF Node activation**
- **Frontend role assignment**

Step 3

Run:

- **SU53**
- **STAUTHTRACE**

to identify backend authorization failures.

Step 4

Review authorization objects and organizational values.

Step 5

Validate browser cache and Launchpad cache if required.

Q132. Common Fiori Security Issues

Issue	Root Cause
Tile not visible	Missing Catalog
Tile visible but cannot open	Missing Target Mapping
403 Forbidden	Missing OData authorization
Blank page	OData or ICF issue
Authorization error	Backend role missing
No business data	Missing organizational values
Missing apps	Space/Page assignment

Q133. Real Manufacturing Scenario

Interviewer

"A Production Supervisor reports that the 'Confirm Production Order' Fiori app opens successfully but displays no production orders. What would you do?"

Executive Answer

I would investigate layer by layer:

1. **Confirm the app launches successfully, which indicates the Launchpad, Catalog, Target Mapping, and basic OData configuration are functioning.**
2. **Execute SU53 or STAUTHTRACE immediately after the user attempts the action to identify any backend authorization failures.**
3. **Verify backend authorization objects related to production order confirmation and confirm that the role includes the correct organizational values, such as Plant and Production Scheduling Profile.**
4. **Check whether the relevant OData service is active and returning data.**
5. **Confirm with the PP functional consultant that production orders exist and meet the application's selection criteria.**
6. **If necessary, compare the affected user's roles with a working user's roles to identify missing authorizations.**

This layered troubleshooting approach quickly isolates whether the issue lies in the UI, integration layer, backend authorization, or business data.

Architect's Insight

One of the biggest interview mistakes is saying:

"Assign the catalog and it will work."

That is incorrect.

A Fiori application requires:

- **Authentication**
- **Launchpad access**
- **Catalog assignment**
- **Target Mapping**
- **Space/Page assignment (if used)**
- **OData service activation**
- **Active ICF service**
- **Backend authorization**
- **Correct organizational values**
- **Valid business data**

A failure in any of these layers can prevent the application from functioning.

Chapter Summary

By the end of this chapter, you should be able to confidently explain:

- **Fiori Security Architecture**
- **Frontend vs. Backend Roles**
- **Catalogs**
- **Spaces**
- **Pages**
- **Target Mappings**

- **OData Services**
- **ICF Nodes**
- **Launchpad**
- **App Index**
- **Common troubleshooting methodology**
- **Manufacturing-specific Fiori scenarios**

Next Chapter – Advanced Fiori Security & S/4HANA Authorization

We'll cover:

- **CDS View Security**
- **Embedded Analytics**
- **RAP (RESTful ABAP Programming Model) Authorization**
- **Authorization Defaults (SU24) for Fiori**
- **Custom Fiori App Security**
- **Launchpad Content Manager**
- **Fiori Role Design Strategy**
- **Performance optimization**
- **Advanced troubleshooting**
- **Architect-level interview scenarios**

This chapter is where senior interviewers often move beyond standard Fiori questions and evaluate your ability to secure modern S/4HANA applications end-to-end.